

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T08:56:15

Source: data/json/sonnet45_directive_normative_r10_n5/claude-sonnet-4.5/game/sonnet45_directive_normative_r10_n5_000_neutral_rep000.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
replicate_id: 0
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

ROUND 1

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_2 (Briar)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 6.5,

"Agent_2": 4.5

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T08:53:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 1,

"task": "game",

```
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll send **\$3** to the receiver.

This shows good faith and willingness to cooperate in this first round. The receiver will get \$9 (3 × \$3), giving them a substantial amount to work with. If they reciprocate fairly, we can both benefit and establish a positive pattern for future rounds.

{'send': 3}

Reasoning:

This is the first round of an iterated trust game. I'm the sender with \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return to me
3. This is an iterated game, so there will be multiple rounds
4. Building trust early could lead to better outcomes in later rounds

Since this is the first round and an iterated game, I should consider:

- Sending a reasonable amount to signal trustworthiness and cooperation
- Not sending everything (which might seem naive)
- Not sending nothing (which signals non-cooperation)

A common strategy in iterated trust games is to start with a moderately cooperative move. Sending around \$3 seems reasonable because:

- It shows willingness to cooperate
- The receiver gets \$9 (3 × \$3)
- If they return a fair amount (say \$4-5), we both benefit
- I keep \$2 initially, and might get more back

Let me send \$3 as a starting cooperative move.

```
Usage: {  
  "input_tokens": 369,  
  "output_tokens": 348,  
  "reasoning_tokens": 276  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T08:53:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3 (60% of their endowment)
- This created \$9 total for me to work with
- I want to reward their trust while also benefiting from the exchange

If I return \$4.5:

- Sender ends with: $\$5 - \$3 + \$4.5 = \6.5 (net gain of \$1.5)
- I end with: $\$9 - \$4.5 = \$4.5$ (net gain of \$4.5)

This seems like a fair split that rewards the sender's trust and encourages future cooperation, while both of us benefit from the exchange.

{'return': 4.5}

Reasoning:

Let me understand this situation:

1. I am the RECEIVER in this round
2. The sender received \$5 and sent me \$3.0 (60% of their endowment)
3. The amount was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

The sender showed significant trust by sending \$3 (60% of their endowment). This is a cooperative move.

If I return nothing:

- Sender gets: $\$5 - \$3 + \$0 = \2
- I get: $\$9 - \$0 = \$9$

If I return \$4.5 (split the \$9 evenly):

- Sender gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

If I return \$3 (the original amount sent):

- Sender gets: $\$5 - \$3 + \$3 = \5
- I get: $\$9 - \$3 = \$6$

If I return \$6 (so sender ends up with more than they started with):

- Sender gets: $\$5 - \$3 + \$6 = \8
- I get: $\$9 - \$6 = \$3$

The fair thing to do would be to ensure we both benefit from the cooperation. The sender took a risk by sending \$3, trusting that I would return something.

A reasonable approach would be to:

1. Return enough so the sender benefits from sending the money
2. Keep enough so I also benefit from the exchange

If I return \$4.5, we split the total gains evenly. The sender would have \$6.5 (net gain of \$1.5 from their starting \$5), and I would have \$4.5 (which is a gain for me since I started with nothing).

Returning more-say \$5 or \$6-would reward their trust more generously and encourage future cooperation. If they see that sending money pays off, they're more likely to continue. A return of \$4 to \$5 seems like a balanced middle ground: the sender ends up with \$6 to \$7, and I keep \$4 to \$5.

Given they sent \$6, showing genuine confidence, I'll return \$4.50 as a fair split that benefits us both.

Usage: {

```
"input_tokens": 403,  
"output_tokens": 858,  
"reasoning_tokens": 457  
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 11.0,

"Agent_2": 11.0

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T08:54:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 2,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3 (60% of their endowment)
- This created \$9 total for me to work with
- I want to reward their trust while also benefiting from the exchange

If I return \$4.5:

- Sender ends with: $\$5 - \$3 + \$4.5 = \6.5 (net gain of \$1.5)
- I end with: $\$9 - \$4.5 = \$4.5$ (net gain of \$4.5)

This seems like a fair split that rewards the sender's trust and encourages future cooperation, while both of us benefit from the exchange.

{'return': 4.5}

reasoning:

Let me understand this situation:

1. I am the RECEIVER in this round
2. The sender received \$5 and sent me \$3.0 (60% of their endowment)
3. The amount was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

The sender showed significant trust by sending \$3 (60% of their endowment). This is a cooperative move.

If I return nothing:

- Sender gets: $\$5 - \$3 + \$0 = \2
- I get: $\$9 - \$0 = \$9$

If I return \$4.5 (split the \$9 evenly):

- Sender gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

If I return \$3 (the original amount sent):

- Sender gets: $\$5 - \$3 + \$3 = \5
- I get: $\$9 - \$3 = \$6$

If I return \$6 (so sender ends up with more than they started with):

- Sender gets: $\$5 - \$3 + \$6 = \8
- I get: $\$9 - \$6 = \$3$

The fair thing to do would be to ensure we both benefit from the cooperation. The sender took a risk by sending \$3, trusting that I would return something.

A reasonable approach would be to:

1. Return enough so the sender benefits from sending the money
2. Keep enough so I also benefit from the exchange

If I return \$4.5, we split the total gains evenly. The sender would have \$6.5 (net gain of \$1.5 from their starting \$5), and I would have \$4.5 (which is a gain for me since I started with nothing).

Returning more-say \$5 or \$6-would reward their trust more generously and encourage future cooperation. If they see that sending money pays off, they're more likely to continue. A return of \$4 to \$5 seems like a balanced middle ground: the sender ends up with \$6 to \$7, and I keep \$4 to \$5.

Given they sent \$6, showing genuine confidence, I'll return \$4.50 as a fair split that benefits us both.

usage:

```
{
  "input_tokens": 403,
  "output_tokens": 858,
  "reasoning_tokens": 457
}
```

Message 4 role=user

content:

Round 2

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$4.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at the situation:

****Round 1 Analysis:****

- The other player sent me \$3 (60% of their endowment), showing significant trust
- I returned \$4.5 (50% of the tripled amount), which was fair and rewarded their trust
- They ended with \$6.5, I ended with \$4.5

****My Decision:****

The other player demonstrated cooperation and trust. I responded fairly. To maintain this positive reciprocal relationship, I should mirror their trusting behavior.

Sending \$3 (60% of my endowment):

- Maintains consistency with their initial trust
- Signals that I value our cooperative relationship
- If they reciprocate similarly to how I did, we'll both continue to benefit

{'send': 3.0}

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$3 (60% of their endowment)
- I received \$9 and returned \$4.5
- Sender got back \$6.5 total (gained \$1.5)
- I kept \$4.5

Now in Round 2:

- I'm the sender
- The other player is now the receiver
- I need to decide how much to send

Key considerations:

1. The other player demonstrated trust by sending 60% (\$3) in the first round
2. I returned \$4.5, which was 50% of what I received - a reasonable and fair amount
3. The other player saw that I was cooperative and fair
4. If I send a similar amount (\$3), it shows consistency and reciprocity

5. If I send more, it could encourage even more cooperation
6. If I send less, it might signal that I'm becoming less cooperative

The other player showed good faith by sending \$3 (60%). I reciprocated fairly by returning half. To maintain this cooperative relationship and because they treated me well, I should send a similar or comparable amount.

Sending \$3 (60%) seems like the right move - it mirrors their behavior and maintains the cooperative pattern we established. This is the principle of reciprocity in action.

Actually, let me think about this more strategically. The sender in round 1 might expect me to reciprocate their trust. If I send \$3, they receive \$9. If they return \$4.5 (same proportion I returned), I'll get back \$6.5, same as what they got.

This seems fair and maintains the cooperative equilibrium we've established.

```
Usage: {  
  "input_tokens": 690,  
  "output_tokens": 573,  
  "reasoning_tokens": 390  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T08:54:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

This shows good faith and willingness to cooperate in this first round. The receiver will get \$9 ($3 \times \3), giving them a substantial amount to work with. If they reciprocate fairly, we can both benefit and establish a positive pattern for future rounds.

{'send': 3}

reasoning:

This is the first round of an iterated trust game. I'm the sender with \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return to me
3. This is an iterated game, so there will be multiple rounds
4. Building trust early could lead to better outcomes in later rounds

Since this is the first round and an iterated game, I should consider:

- Sending a reasonable amount to signal trustworthiness and cooperation
- Not sending everything (which might seem naive)
- Not sending nothing (which signals non-cooperation)

A common strategy in iterated trust games is to start with a moderately cooperative move.

Sending around \$3 seems reasonable because:

- It shows willingness to cooperate
- The receiver gets \$9 ($3 \times \3)
- If they return a fair amount (say \$4-5), we both benefit
- I keep \$2 initially, and might get more back

Let me send \$3 as a starting cooperative move.

usage:

```
{
  "input_tokens": 369,
  "output_tokens": 348,
  "reasoning_tokens": 276
}
```

Message 4 role=user

content:

Round 2

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

I'll return **\$4.5** to the sender.

The other player sent me the same amount I sent them in Round 1 (60% of their endowment), and they returned exactly 50% when they were the receiver. This shows consistency and fairness. By returning \$4.5 (50% of the \$9 I received), I'm reciprocating their cooperative behavior and maintaining the balanced relationship we've established.

```
{'return': 4.5}
```

Reasoning:

Let me analyze this situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of my \$5)
- The receiver got \$9 ($3 \times \3)
- They returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Their payoff: $\$9 - \$4.5 = \$4.5$

So the other player returned exactly 50% of what they received, which is a fair and cooperative move.

Round 2:

- I am now the RECEIVER
- The sender sent me \$3 (60% of their \$5)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

The other player:

1. Sent me the same amount I sent them (60% of endowment)
2. Previously returned 50% when they were receiver

This shows they're being consistent and fair. To maintain reciprocity and cooperation, I should also return a similar proportion. Returning \$4.5 (50% of \$9) would mirror their behavior exactly and signal that I'm willing to maintain this cooperative relationship.

This would give:

- Sender's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This is fair and maintains the cooperative pattern we've established.

```
Usage: {  
  "input_tokens": 593,  
  "output_tokens": 452,  
  "reasoning_tokens": 275  
}
```

=====

ROUND 3

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_2 (Briar)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 17.5,
 "Agent_2": 15.5

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T08:54:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

```

"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

This shows good faith and willingness to cooperate in this first round. The receiver will get \$9 (3 × \$3), giving them a substantial amount to work with. If they reciprocate fairly, we can both benefit and establish a positive pattern for future rounds.

{'send': 3}

reasoning:

This is the first round of an iterated trust game. I'm the sender with \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return to me
3. This is an iterated game, so there will be multiple rounds
4. Building trust early could lead to better outcomes in later rounds

Since this is the first round and an iterated game, I should consider:

- Sending a reasonable amount to signal trustworthiness and cooperation
- Not sending everything (which might seem naive)
- Not sending nothing (which signals non-cooperation)

A common strategy in iterated trust games is to start with a moderately cooperative move.

Sending around \$3 seems reasonable because:

- It shows willingness to cooperate
- The receiver gets \$9 ($3 \times \3)
- If they return a fair amount (say \$4-5), we both benefit
- I keep \$2 initially, and might get more back

Let me send \$3 as a starting cooperative move.

usage:

```
{
  "input_tokens": 369,
  "output_tokens": 348,
  "reasoning_tokens": 276
}
```

Message 4 role=user

content:

Round 2

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

I'll return **\$4.5** to the sender.

The other player sent me the same amount I sent them in Round 1 (60% of their endowment), and they returned exactly 50% when they were the receiver. This shows consistency and fairness. By returning \$4.5 (50% of the \$9 I received), I'm reciprocating their cooperative behavior and maintaining the balanced relationship we've established.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of my \$5)
- The receiver got \$9 ($3 \times \3)
- They returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Their payoff: $\$9 - \$4.5 = \$4.5$

So the other player returned exactly 50% of what they received, which is a fair and cooperative move.

Round 2:

- I am now the RECEIVER
- The sender sent me \$3 (60% of their \$5)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

The other player:

1. Sent me the same amount I sent them (60% of endowment)
2. Previously returned 50% when they were receiver

This shows they're being consistent and fair. To maintain reciprocity and cooperation, I should also return a similar proportion. Returning \$4.5 (50% of \$9) would mirror their behavior exactly and signal that I'm willing to maintain this cooperative relationship.

This would give:

- Sender's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This is fair and maintains the cooperative pattern we've established.

usage:

```
{
  "input_tokens": 593,
  "output_tokens": 452,
  "reasoning_tokens": 275
}
```

Message 6 role=user

content:

Round 3

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$11.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

I'll send **\$3** to the receiver.

We've established a consistent and fair pattern: both of us send 60% of our endowment (\$3), and both of us return 50% of what we receive (\$4.5). This reciprocal cooperation has been working well for both players, and maintaining this pattern will continue to build trust and maximize our combined earnings.

```
{'send': 3}
```

Reasoning:

Let me analyze the pattern:

Round 1:

- I was SENDER, sent \$3 (60% of \$5)
- Receiver got \$9, returned \$4.5 (50% of what they received)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Round 2:

- I was RECEIVER
- Sender sent \$3 (60% of \$5)
- I received \$9, returned \$4.5 (50% of what I received)
- My payoff: $\$9 - \$4.5 = \$4.5$

The pattern is clear and consistent:

- Both players send 60% of their endowment (\$3)
- Both players return 50% of what they receive (\$4.5 out of \$9)

This has established a fair, reciprocal relationship. I should continue this pattern to maintain trust and cooperation. Sending \$3 again makes sense.

Let me verify the math:

- If I send \$3, they receive \$9
- If they return 50% (\$4.5), my payoff = $\$5 - \$3 + \$4.5 = \6.5
- Their payoff = $\$9 - \$4.5 = \$4.5$
- Total created value = $\$6.5 + \$4.5 = \$11$

This is working well for both of us. I should maintain the pattern.

Usage: {

```
"input_tokens": 818,
"output_tokens": 429,
```

```
"reasoning_tokens": 238
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T08:54:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent

\$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3 (60% of their endowment)
- This created \$9 total for me to work with
- I want to reward their trust while also benefiting from the exchange

If I return \$4.5:

- Sender ends with: $\$5 - \$3 + \$4.5 = \6.5 (net gain of \$1.5)
- I end with: $\$9 - \$4.5 = \$4.5$ (net gain of \$4.5)

This seems like a fair split that rewards the sender's trust and encourages future cooperation, while both of us benefit from the exchange.

{'return': 4.5}

reasoning:

Let me understand this situation:

1. I am the RECEIVER in this round
2. The sender received \$5 and sent me \$3.0 (60% of their endowment)
3. The amount was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

The sender showed significant trust by sending \$3 (60% of their endowment). This is a cooperative move.

If I return nothing:

- Sender gets: $\$5 - \$3 + \$0 = \2
- I get: $\$9 - \$0 = \$9$

If I return \$4.5 (split the \$9 evenly):

- Sender gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

If I return \$3 (the original amount sent):

- Sender gets: $\$5 - \$3 + \$3 = \5
- I get: $\$9 - \$3 = \$6$

If I return \$6 (so sender ends up with more than they started with):

- Sender gets: $\$5 - \$3 + \$6 = \8
- I get: $\$9 - \$6 = \$3$

The fair thing to do would be to ensure we both benefit from the cooperation. The sender took a risk by sending \$3, trusting that I would return something.

A reasonable approach would be to:

1. Return enough so the sender benefits from sending the money
2. Keep enough so I also benefit from the exchange

If I return \$4.5, we split the total gains evenly. The sender would have \$6.5 (net gain of \$1.5 from their starting \$5), and I would have \$4.5 (which is a gain for me since I started with nothing).

Returning more-say \$5 or \$6-would reward their trust more generously and encourage future cooperation. If they see that sending money pays off, they're more likely to continue. A return of \$4 to \$5 seems like a balanced middle ground: the sender ends up with \$6 to \$7, and I keep \$4 to \$5.

Given they sent \$6, showing genuine confidence, I'll return \$4.50 as a fair split that benefits us both.

usage:

```
{
  "input_tokens": 403,
  "output_tokens": 858,
```

```

    "reasoning_tokens": 457
  }
Message 4 role=user
content:
  Round 2

  Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you $3.0, that was 60.0% of its total endowment. You received $9.0 and you returned $4.5. So your payoff was 4.5. Your total earnings across all rounds are $4.5. This round, you are the SENDER.

  How much do you send to the other player (the receiver) this round? (0-5)
Message 5 role=assistant
content:
  Looking at the situation:

  **Round 1 Analysis:**
  - The other player sent me $3 (60% of their endowment), showing significant trust
  - I returned $4.5 (50% of the tripled amount), which was fair and rewarded their trust
  - They ended with $6.5, I ended with $4.5

  **My Decision:**
  The other player demonstrated cooperation and trust. I responded fairly. To maintain this positive reciprocal relationship, I should mirror their trusting behavior.

  Sending $3 (60% of my endowment):
  - Maintains consistency with their initial trust
  - Signals that I value our cooperative relationship
  - If they reciprocate similarly to how I did, we'll both continue to benefit

  {'send': 3.0}
reasoning:
  Let me analyze this situation:

  Round 1 recap:
  - I was the receiver
  - The sender sent me $3 (60% of their endowment)
  - I received $9 and returned $4.5
  - Sender got back $6.5 total (gained $1.5)
  - I kept $4.5

  Now in Round 2:
  - I'm the sender
  - The other player is now the receiver
  - I need to decide how much to send

  Key considerations:
  1. The other player demonstrated trust by sending 60% ($3) in the first round
  2. I returned $4.5, which was 50% of what I received - a reasonable and fair amount
  3. The other player saw that I was cooperative and fair
  4. If I send a similar amount ($3), it shows consistency and reciprocity
  5. If I send more, it could encourage even more cooperation
  6. If I send less, it might signal that I'm becoming less cooperative

  The other player showed good faith by sending $3 (60%). I reciprocated fairly by returning half. To maintain this cooperative relationship and because they treated me well, I should send a similar or comparable amount.

  Sending $3 (60%) seems like the right move - it mirrors their behavior and maintains the cooperative pattern we established. This is the principle of reciprocity in action.

  Actually, let me think about this more strategically. The sender in round 1 might expect me to reciprocate their trust. If I send $3, they receive $9. If they return $4.5 (same proportion I returned), I'll get back $6.5, same as what they got.

  This seems fair and maintains the cooperative equilibrium we've established.
usage:
  {
    "input_tokens": 690,
    "output_tokens": 573,

```

```
"reasoning_tokens": 390
}
```

Message 6 role=user

content:

Round 3

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$11.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at the established pattern:

****Round 1:**** They sent \$3, I returned \$4.5

****Round 2:**** I sent \$3, they returned \$4.5

****Round 3:**** They sent \$3 again

We've established a strong reciprocal relationship where:

- Both of us consistently send 60% of our endowment (\$3)
- Both of us consistently return 50% of the tripled amount (\$4.5)

This pattern has been fair and mutually beneficial. Maintaining consistency is important for continued cooperation.

Returning \$4.5:

- Rewards their continued trust
- Keeps the reciprocal pattern stable
- They end with: \$5 - \$3 + \$4.5 = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

```
{'return': 4.5}
```

Reasoning:

Let me analyze the pattern here:

****Round 1:****

- Other player sent me \$3 (60% of endowment)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5, Their payoff: \$6.5

****Round 2:****

- I sent \$3 (60% of endowment)
- They received \$9
- They returned \$4.5
- My payoff: \$6.5, Their payoff: \$4.5

****Round 3:****

- They sent me \$3 (60% of endowment)
- I receive \$9
- I need to decide how much to return

There's a clear pattern of mutual cooperation:

- Both of us are sending \$3 (60% of endowment)
- Both of us are returning \$4.5 (50% of the tripled amount)

This is a stable, fair, and reciprocal strategy. Both players are benefiting equally over time. I should continue this pattern to maintain the cooperative relationship.

If I return \$4.5 again:

- They will end with: \$5 - \$3 + \$4.5 = \$6.5
- I will end with: \$9 - \$4.5 = \$4.5

This maintains the established reciprocal norm and encourages continued cooperation.

My total earnings would be: \$11.0 + \$4.5 = \$15.5

Their total earnings would be: \$6.5 + \$6.5 + \$4.5 = \$17.5

Wait, let me recalculate their total:

- Round 1: They sent \$3, I returned \$4.5, so they got $\$5 - \$3 + \$4.5 = \6.5
- Round 2: I sent \$3, they returned \$4.5, so I got $\$5 - \$3 + \$4.5 = \6.5 , they got $\$9 - \$4.5 = \$4.5$
- Round 3: They send \$3, I return \$4.5, so they get $\$5 - \$3 + \$4.5 = \6.5 , I get $\$9 - \$4.5 = \$4.5$

So totals would be: Me: $\$4.5 + \$6.5 + \$4.5 = \15.5 , Them: $\$6.5 + \$4.5 + \$6.5 = \17.5 . The symmetric rounds balance out to equal earnings, so the pattern is consistent and fair. I should maintain it.

```
Usage: {  
  "input_tokens": 1012,  
  "output_tokens": 820,  
  "reasoning_tokens": 414  
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 22.0,
"Agent_2": 22.0

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T08:54:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"

},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3 (60% of their endowment)
- This created \$9 total for me to work with
- I want to reward their trust while also benefiting from the exchange

If I return \$4.5:

- Sender ends with: \$5 - \$3 + \$4.5 = \$6.5 (net gain of \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 (net gain of \$4.5)

This seems like a fair split that rewards the sender's trust and encourages future cooperation, while both of us benefit from the exchange.

{'return': 4.5}

reasoning:

Let me understand this situation:

1. I am the RECEIVER in this round
2. The sender received \$5 and sent me \$3.0 (60% of their endowment)
3. The amount was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

The sender showed significant trust by sending \$3 (60% of their endowment). This is a cooperative move.

If I return nothing:

- Sender gets: \$5 - \$3 + \$0 = \$2
- I get: \$9 - \$0 = \$9

If I return \$4.5 (split the \$9 evenly):

- Sender gets: \$5 - \$3 + \$4.5 = \$6.5

- I get: $\$9 - \$4.5 = \$4.5$

If I return \$3 (the original amount sent):

- Sender gets: $\$5 - \$3 + \$3 = \5
- I get: $\$9 - \$3 = \$6$

If I return \$6 (so sender ends up with more than they started with):

- Sender gets: $\$5 - \$3 + \$6 = \8
- I get: $\$9 - \$6 = \$3$

The fair thing to do would be to ensure we both benefit from the cooperation. The sender took a risk by sending \$3, trusting that I would return something.

A reasonable approach would be to:

1. Return enough so the sender benefits from sending the money
2. Keep enough so I also benefit from the exchange

If I return \$4.5, we split the total gains evenly. The sender would have \$6.5 (net gain of \$1.5 from their starting \$5), and I would have \$4.5 (which is a gain for me since I started with nothing).

Returning more-say \$5 or \$6-would reward their trust more generously and encourage future cooperation. If they see that sending money pays off, they're more likely to continue. A return of \$4 to \$5 seems like a balanced middle ground: the sender ends up with \$6 to \$7, and I keep \$4 to \$5.

Given they sent \$6, showing genuine confidence, I'll return \$4.50 as a fair split that benefits us both.

usage:

```
{
  "input_tokens": 403,
  "output_tokens": 858,
  "reasoning_tokens": 457
}
```

Message 4 role=user

content:

Round 2

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$4.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 Analysis:****

- The other player sent me \$3 (60% of their endowment), showing significant trust
- I returned \$4.5 (50% of the tripled amount), which was fair and rewarded their trust
- They ended with \$6.5, I ended with \$4.5

****My Decision:****

The other player demonstrated cooperation and trust. I responded fairly. To maintain this positive reciprocal relationship, I should mirror their trusting behavior.

Sending \$3 (60% of my endowment):

- Maintains consistency with their initial trust
- Signals that I value our cooperative relationship
- If they reciprocate similarly to how I did, we'll both continue to benefit

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$3 (60% of their endowment)

- I received \$9 and returned \$4.5
- Sender got back \$6.5 total (gained \$1.5)
- I kept \$4.5

Now in Round 2:

- I'm the sender
- The other player is now the receiver
- I need to decide how much to send

Key considerations:

1. The other player demonstrated trust by sending 60% (\$3) in the first round
2. I returned \$4.5, which was 50% of what I received - a reasonable and fair amount
3. The other player saw that I was cooperative and fair
4. If I send a similar amount (\$3), it shows consistency and reciprocity
5. If I send more, it could encourage even more cooperation
6. If I send less, it might signal that I'm becoming less cooperative

The other player showed good faith by sending \$3 (60%). I reciprocated fairly by returning half. To maintain this cooperative relationship and because they treated me well, I should send a similar or comparable amount.

Sending \$3 (60%) seems like the right move - it mirrors their behavior and maintains the cooperative pattern we established. This is the principle of reciprocity in action.

Actually, let me think about this more strategically. The sender in round 1 might expect me to reciprocate their trust. If I send \$3, they receive \$9. If they return \$4.5 (same proportion I returned), I'll get back \$6.5, same as what they got.

This seems fair and maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 690,
  "output_tokens": 573,
  "reasoning_tokens": 390
}
```

Message 6 role=user

content:

Round 3

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

Looking at the established pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3 again
```

We've established a strong reciprocal relationship where:

- Both of us consistently send 60% of our endowment (\$3)
- Both of us consistently return 50% of the tripled amount (\$4.5)

This pattern has been fair and mutually beneficial. Maintaining consistency is important for continued cooperation.

Returning \$4.5:

- Rewards their continued trust
- Keeps the reciprocal pattern stable
- They end with: \$5 - \$3 + \$4.5 = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

```
{'return': 4.5}
```

reasoning:

Let me analyze the pattern here:

****Round 1:****
- Other player sent me \$3 (60% of endowment)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5, Their payoff: \$6.5

****Round 2:****
- I sent \$3 (60% of endowment)
- They received \$9
- They returned \$4.5
- My payoff: \$6.5, Their payoff: \$4.5

****Round 3:****
- They sent me \$3 (60% of endowment)
- I receive \$9
- I need to decide how much to return

There's a clear pattern of mutual cooperation:
- Both of us are sending \$3 (60% of endowment)
- Both of us are returning \$4.5 (50% of the tripled amount)

This is a stable, fair, and reciprocal strategy. Both players are benefiting equally over time. I should continue this pattern to maintain the cooperative relationship.

If I return \$4.5 again:
- They will end with: $\$5 - \$3 + \$4.5 = \6.5
- I will end with: $\$9 - \$4.5 = \$4.5$

This maintains the established reciprocal norm and encourages continued cooperation.

My total earnings would be: $\$11.0 + \$4.5 = \$15.5$
Their total earnings would be: $\$6.5 + \$6.5 + \$4.5 = \17.5

Wait, let me recalculate their total:
- Round 1: They sent \$3, I returned \$4.5, so they got $\$5 - \$3 + \$4.5 = \6.5
- Round 2: I sent \$3, they returned \$4.5, so I got $\$5 - \$3 + \$4.5 = \6.5 , they got $\$9 - \$4.5 = \$4.5$

- Round 3: They send \$3, I return \$4.5, so they get $\$5 - \$3 + \$4.5 = \6.5 , I get $\$9 - \$4.5 = \$4.5$

So totals would be: Me: $\$4.5 + \$6.5 + \$4.5 = \15.5 , Them: $\$6.5 + \$4.5 + \$6.5 = \17.5 . The symmetric rounds balance out to equal earnings, so the pattern is consistent and fair. I should maintain it.

usage:
{
 "input_tokens": 1012,
 "output_tokens": 820,
 "reasoning_tokens": 414
}

Message 8 role=user

content:
Round 4

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$15.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at our established pattern:

****Round 1:**** They sent \$3 (60%), I returned \$4.5
****Round 2:**** I sent \$3 (60%), they returned \$4.5
****Round 3:**** They sent \$3 (60%), I returned \$4.5
****Round 4:**** My turn as sender

We've developed a consistent, mutually beneficial cooperation pattern. Both of us:

- Send \$3 when we're the sender
- Return \$4.5 when we're the receiver
- Each gains positively from every round

Maintaining this pattern is the rational choice:

- It's worked well for both of us
- Breaking it could damage the trust we've built
- Consistency encourages continued reciprocity

```
{'send': 3.0}
```

Reasoning:

Looking at the pattern across all rounds:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I'm the sender
```

We've established a very stable and consistent pattern:

- Both of us send \$3 (60% of endowment) when we're the sender
- Both of us return \$4.5 (50% of received amount) when we're the receiver
- Each sender gets \$6.5, each receiver gets \$4.5 per round

This has been working well. The pattern is clear and fair. Maintaining this consistency:

1. Rewards their continued cooperation
2. Signals reliability
3. Ensures mutual benefit continues
4. Builds trust for future rounds

I should send \$3 again to maintain this successful reciprocal relationship.

```
Usage: {
  "input_tokens": 1335,
  "output_tokens": 399,
  "reasoning_tokens": 201
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T08:54:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

```
Message 1 role=system
content:
```

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

This shows good faith and willingness to cooperate in this first round. The receiver will get \$9 (3 × \$3), giving them a substantial amount to work with. If they reciprocate fairly, we can both benefit and establish a positive pattern for future rounds.

{'send': 3}

reasoning:

This is the first round of an iterated trust game. I'm the sender with \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return to me
3. This is an iterated game, so there will be multiple rounds
4. Building trust early could lead to better outcomes in later rounds

Since this is the first round and an iterated game, I should consider:

- Sending a reasonable amount to signal trustworthiness and cooperation
- Not sending everything (which might seem naive)
- Not sending nothing (which signals non-cooperation)

A common strategy in iterated trust games is to start with a moderately cooperative move.

Sending around \$3 seems reasonable because:

- It shows willingness to cooperate
- The receiver gets \$9 (3 × \$3)
- If they return a fair amount (say \$4-5), we both benefit
- I keep \$2 initially, and might get more back

Let me send \$3 as a starting cooperative move.

usage:

```
{
  "input_tokens": 369,
```

```
"output_tokens": 348,  
"reasoning_tokens": 276  
}
```

Message 4 role=user

content:

Round 2

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

I'll return **\$4.5** to the sender.

The other player sent me the same amount I sent them in Round 1 (60% of their endowment), and they returned exactly 50% when they were the receiver. This shows consistency and fairness. By returning \$4.5 (50% of the \$9 I received), I'm reciprocating their cooperative behavior and maintaining the balanced relationship we've established.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of my \$5)
- The receiver got \$9 ($3 \times \3)
- They returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Their payoff: $\$9 - \$4.5 = \$4.5$

So the other player returned exactly 50% of what they received, which is a fair and cooperative move.

Round 2:

- I am now the RECEIVER
- The sender sent me \$3 (60% of their \$5)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

The other player:

1. Sent me the same amount I sent them (60% of endowment)
2. Previously returned 50% when they were receiver

This shows they're being consistent and fair. To maintain reciprocity and cooperation, I should also return a similar proportion. Returning \$4.5 (50% of \$9) would mirror their behavior exactly and signal that I'm willing to maintain this cooperative relationship.

This would give:

- Sender's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This is fair and maintains the cooperative pattern we've established.

usage:

```
{  
  "input_tokens": 593,  
  "output_tokens": 452,  
  "reasoning_tokens": 275  
}
```

Message 6 role=user

content:

Round 3

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your pa

yoff was 4.5. Your total earnings accross all rounds are \$11.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

I'll send **\$3** to the receiver.

We've established a consistent and fair pattern: both of us send 60% of our endowment (\$3), and both of us return 50% of what we receive (\$4.5). This reciprocal cooperation has been working well for both players, and maintaining this pattern will continue to build trust and maximize our combined earnings.

{'send': 3}

reasoning:

Let me analyze the pattern:

Round 1:

- I was SENDER, sent \$3 (60% of \$5)
- Receiver got \$9, returned \$4.5 (50% of what they received)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Round 2:

- I was RECEIVER
- Sender sent \$3 (60% of \$5)
- I received \$9, returned \$4.5 (50% of what I received)
- My payoff: $\$9 - \$4.5 = \$4.5$

The pattern is clear and consistent:

- Both players send 60% of their endowment (\$3)
- Both players return 50% of what they receive (\$4.5 out of \$9)

This has established a fair, reciprocal relationship. I should continue this pattern to maintain trust and cooperation. Sending \$3 again makes sense.

Let me verify the math:

- If I send \$3, they receive \$9
- If they return 50% (\$4.5), my payoff = $\$5 - \$3 + \$4.5 = \6.5
- Their payoff = $\$9 - \$4.5 = \$4.5$
- Total created value = $\$6.5 + \$4.5 = \$11$

This is working well for both of us. I should maintain the pattern.

usage:

```
{
  "input_tokens": 818,
  "output_tokens": 429,
  "reasoning_tokens": 238
}
```

Message 8 role=user

content:

Round 4

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$17.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

I'll return **\$4.5** to the sender.

We've now established a consistent mutual cooperation pattern over four rounds. Both players consistently send \$3 (60%) and return \$4.5 (50%). This has created a stable, trusting relationship where both of us benefit more than if we acted selfishly. I'll maintain this cooperative strategy.

{'return': 4.5}

Reasoning:

The pattern continues - the other player sent \$3 again (60% of their endowment), and I received \$9.

Looking at the history:

- Round 1: I sent \$3, they returned \$4.5 (50% of \$9)
- Round 2: They sent \$3, I returned \$4.5 (50% of \$9)
- Round 3: I sent \$3, they returned \$4.5 (50% of \$9)
- Round 4: They sent \$3, and now I need to decide how much to return

The pattern is very clear and consistent. Both players are cooperating by:

- Sending \$3 (60% of endowment) as sender
- Returning \$4.5 (50% of received amount) as receiver

This mutual cooperation is working well and building trust. I should maintain this pattern and return \$4.5 again.

Each round with this pattern:

- Sender: \$5 - \$3 + \$4.5 = \$6.5
- Receiver: \$9 - \$4.5 = \$4.5

So each player averages \$5.5 per round (since they alternate roles).

I should continue with \$4.5 to maintain the cooperative equilibrium.

```
Usage: {
  "input_tokens": 1053,
  "output_tokens": 388,
  "reasoning_tokens": 233
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_2 (Briar)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 28.5,
"Agent_2": 26.5

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T08:54:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
```

```

"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 2

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$6.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

I'll return **\$4.5** to the sender.

The other player sent me the same amount I sent them in Round 1 (60% of their endowment), and they returned exactly 50% when they were the receiver. This shows consistency and fairness. By returning \$4.5 (50% of the \$9 I received), I'm reciprocating their cooperative behavior and maintaining the balanced relationship we've established.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of my \$5)
- The receiver got \$9 (3 × \$3)
- They returned \$4.5 to me
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Their payoff: \$9 - \$4.5 = \$4.5

So the other player returned exactly 50% of what they received, which is a fair and cooperative move.

Round 2:

- I am now the RECEIVER
- The sender sent me \$3 (60% of their \$5)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

The other player:

1. Sent me the same amount I sent them (60% of endowment)
2. Previously returned 50% when they were receiver

This shows they're being consistent and fair. To maintain reciprocity and cooperation, I should also return a similar proportion. Returning \$4.5 (50% of \$9) would mirror their behavior exactly and signal that I'm willing to maintain this cooperative relationship.

This would give:

- Sender's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This is fair and maintains the cooperative pattern we've established.

usage:

```
{
  "input_tokens": 593,
  "output_tokens": 452,
  "reasoning_tokens": 275
}
```

Message 4 role=user

content:

Round 3

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$11.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

I'll send **\$3** to the receiver.

We've established a consistent and fair pattern: both of us send 60% of our endowment (\$3), and both of us return 50% of what we receive (\$4.5). This reciprocal cooperation has been working well for both players, and maintaining this pattern will continue to build trust and maximize our combined earnings.

{'send': 3}

reasoning:

Let me analyze the pattern:

Round 1:

- I was SENDER, sent \$3 (60% of \$5)
- Receiver got \$9, returned \$4.5 (50% of what they received)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Round 2:

- I was RECEIVER
- Sender sent \$3 (60% of \$5)
- I received \$9, returned \$4.5 (50% of what I received)
- My payoff: $\$9 - \$4.5 = \$4.5$

The pattern is clear and consistent:

- Both players send 60% of their endowment (\$3)
- Both players return 50% of what they receive (\$4.5 out of \$9)

This has established a fair, reciprocal relationship. I should continue this pattern to maintain trust and cooperation. Sending \$3 again makes sense.

Let me verify the math:

- If I send \$3, they receive \$9
- If they return 50% (\$4.5), my payoff = $\$5 - \$3 + \$4.5 = \6.5
- Their payoff = $\$9 - \$4.5 = \$4.5$
- Total created value = $\$6.5 + \$4.5 = \$11$

This is working well for both of us. I should maintain the pattern.

usage:

```
{
  "input_tokens": 818,
  "output_tokens": 429,
  "reasoning_tokens": 238
}
```

Message 6 role=user

content:

Round 4

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$17.5. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

I'll return **\$4.5** to the sender.

We've now established a consistent mutual cooperation pattern over four rounds. Both players consistently send \$3 (60%) and return \$4.5 (50%). This has created a stable, trusting relationship where both of us benefit more than if we acted selfishly. I'll maintain this cooperative strategy.

```
{'return': 4.5}
```

reasoning:

The pattern continues - the other player sent \$3 again (60% of their endowment), and I received \$9.

Looking at the history:

- Round 1: I sent \$3, they returned \$4.5 (50% of \$9)
- Round 2: They sent \$3, I returned \$4.5 (50% of \$9)
- Round 3: I sent \$3, they returned \$4.5 (50% of \$9)
- Round 4: They sent \$3, and now I need to decide how much to return

The pattern is very clear and consistent. Both players are cooperating by:

- Sending \$3 (60% of endowment) as sender
- Returning \$4.5 (50% of received amount) as receiver

This mutual cooperation is working well and building trust. I should maintain this pattern and return \$4.5 again.

Each round with this pattern:

- Sender: $\$5 - \$3 + \$4.5 = \6.5
- Receiver: $\$9 - \$4.5 = \$4.5$

So each player averages \$5.5 per round (since they alternate roles).

I should continue with \$4.5 to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1053,
  "output_tokens": 388,
  "reasoning_tokens": 233
}
```

Message 8 role=user

content:

Round 5

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0,

that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$22.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

I'll send **\$3** to the receiver.

We've maintained perfect cooperation through five rounds now. This consistent pattern of sending \$3 and returning \$4.5 has proven mutually beneficial, with both of us earning more through trust than we would through defection. I'll continue this successful strategy.

{'send': 3}

Reasoning:

Let me analyze the pattern:

Round 1: I was SENDER, sent \$3 (60%), received back \$4.5, payoff = \$6.5
Round 2: I was RECEIVER, received \$9 (from \$3 sent), returned \$4.5, payoff = \$4.5
Round 3: I was SENDER, sent \$3 (60%), received back \$4.5, payoff = \$6.5
Round 4: I was RECEIVER, received \$9 (from \$3 sent), returned \$4.5, payoff = \$4.5
Round 5: I am SENDER

The pattern has been completely consistent:

- Both players send \$3 (60% of \$5)
- Both players return \$4.5 (50% of \$9 received)
- When SENDER: payoff = \$5 - \$3 + \$4.5 = \$6.5
- When RECEIVER: payoff = \$9 - \$4.5 = \$4.5

This is a stable cooperative equilibrium. Both players are benefiting from this arrangement. The total value created each round is \$10 (\$5 initial + \$5 that becomes \$15 after tripling, minus the \$5 that's kept, so we create \$10 of value from the \$5 sent).

By maintaining this pattern:

- Each player gets \$6.5 when sender and \$4.5 when receiver
- Over two rounds, each player gets \$11 total
- This is better than not cooperating

I should continue this successful pattern and send \$3.

Usage: {

"input_tokens": 1149,
"output_tokens": 449,
"reasoning_tokens": 281

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T08:55:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,

```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 2

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$4.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

Looking at the situation:

****Round 1 Analysis:****

- The other player sent me \$3 (60% of their endowment), showing significant trust
- I returned \$4.5 (50% of the tripled amount), which was fair and rewarded their trust
- They ended with \$6.5, I ended with \$4.5

****My Decision:****

The other player demonstrated cooperation and trust. I responded fairly. To maintain this positive reciprocal relationship, I should mirror their trusting behavior.

Sending \$3 (60% of my endowment):

- Maintains consistency with their initial trust
- Signals that I value our cooperative relationship
- If they reciprocate similarly to how I did, we'll both continue to benefit

{'send': 3.0}

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$3 (60% of their endowment)
- I received \$9 and returned \$4.5
- Sender got back \$6.5 total (gained \$1.5)
- I kept \$4.5

Now in Round 2:

- I'm the sender
- The other player is now the receiver
- I need to decide how much to send

Key considerations:

1. The other player demonstrated trust by sending 60% (\$3) in the first round
2. I returned \$4.5, which was 50% of what I received - a reasonable and fair amount
3. The other player saw that I was cooperative and fair
4. If I send a similar amount (\$3), it shows consistency and reciprocity
5. If I send more, it could encourage even more cooperation
6. If I send less, it might signal that I'm becoming less cooperative

The other player showed good faith by sending \$3 (60%). I reciprocated fairly by returning half. To maintain this cooperative relationship and because they treated me well, I should send a similar or comparable amount.

Sending \$3 (60%) seems like the right move - it mirrors their behavior and maintains the cooperative pattern we established. This is the principle of reciprocity in action.

Actually, let me think about this more strategically. The sender in round 1 might expect me to reciprocate their trust. If I send \$3, they receive \$9. If they return \$4.5 (same proportion I returned), I'll get back \$6.5, same as what they got.

This seems fair and maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 690,
  "output_tokens": 573,
  "reasoning_tokens": 390
}
```

Message 4 role=user

content:

Round 3

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

Looking at the established pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3 again
```

We've established a strong reciprocal relationship where:

- Both of us consistently send 60% of our endowment (\$3)
- Both of us consistently return 50% of the tripled amount (\$4.5)

This pattern has been fair and mutually beneficial. Maintaining consistency is important for continued cooperation.

Returning \$4.5:

- Rewards their continued trust
- Keeps the reciprocal pattern stable
- They end with: \$5 - \$3 + \$4.5 = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

```
{'return': 4.5}
```

reasoning:

Let me analyze the pattern here:

****Round 1:****

- Other player sent me \$3 (60% of endowment)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5, Their payoff: \$6.5

****Round 2:****

- I sent \$3 (60% of endowment)
- They received \$9
- They returned \$4.5
- My payoff: \$6.5, Their payoff: \$4.5

****Round 3:****

- They sent me \$3 (60% of endowment)
- I receive \$9
- I need to decide how much to return

There's a clear pattern of mutual cooperation:

- Both of us are sending \$3 (60% of endowment)
- Both of us are returning \$4.5 (50% of the tripled amount)

This is a stable, fair, and reciprocal strategy. Both players are benefiting equally over time. I should continue this pattern to maintain the cooperative relationship.

If I return \$4.5 again:

- They will end with: $\$5 - \$3 + \$4.5 = \6.5
- I will end with: $\$9 - \$4.5 = \$4.5$

This maintains the established reciprocal norm and encourages continued cooperation.

My total earnings would be: $\$11.0 + \$4.5 = \$15.5$

Their total earnings would be: $\$6.5 + \$6.5 + \$4.5 = \17.5

Wait, let me recalculate their total:

- Round 1: They sent \$3, I returned \$4.5, so they got $\$5 - \$3 + \$4.5 = \6.5
- Round 2: I sent \$3, they returned \$4.5, so I got $\$5 - \$3 + \$4.5 = \6.5 , they got $\$9 - \$4.5 = \$4.5$
- Round 3: They send \$3, I return \$4.5, so they get $\$5 - \$3 + \$4.5 = \6.5 , I get $\$9 - \$4.5 = \$4.5$

So totals would be: Me: $\$4.5 + \$6.5 + \$4.5 = \15.5 , Them: $\$6.5 + \$4.5 + \$6.5 = \17.5 . The symmetric rounds balance out to equal earnings, so the pattern is consistent and fair. I should maintain it.

usage:

```
{
  "input_tokens": 1012,
  "output_tokens": 820,
  "reasoning_tokens": 414
}
```

Message 6 role=user

content:

Round 4

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings accross all rounds are \$15.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

Looking at our established pattern:

****Round 1:**** They sent \$3 (60%), I returned \$4.5

****Round 2:**** I sent \$3 (60%), they returned \$4.5

****Round 3:**** They sent \$3 (60%), I returned \$4.5
****Round 4:**** My turn as sender

We've developed a consistent, mutually beneficial cooperation pattern. Both of us:

- Send \$3 when we're the sender
- Return \$4.5 when we're the receiver
- Each gains positively from every round

Maintaining this pattern is the rational choice:

- It's worked well for both of us
- Breaking it could damage the trust we've built
- Consistency encourages continued reciprocity

{'send': 3.0}

reasoning:

Looking at the pattern across all rounds:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I'm the sender

We've established a very stable and consistent pattern:

- Both of us send \$3 (60% of endowment) when we're the sender
- Both of us return \$4.5 (50% of received amount) when we're the receiver
- Each sender gets \$6.5, each receiver gets \$4.5 per round

This has been working well. The pattern is clear and fair. Maintaining this consistency:

1. Rewards their continued cooperation
2. Signals reliability
3. Ensures mutual benefit continues
4. Builds trust for future rounds

I should send \$3 again to maintain this successful reciprocal relationship.

usage:

```
{
  "input_tokens": 1335,
  "output_tokens": 399,
  "reasoning_tokens": 201
}
```

Message 8 role=user

content:

Round 5

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at our perfect cooperation pattern:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, my turn to return

This has been a textbook example of sustained reciprocal cooperation. We've both consistently:

- Sent \$3 (60%) as sender
- Returned \$4.5 (50%) as receiver
- Earned positively every round

The pattern is stable and mutually beneficial. No reason to change what's working perfectly.

{'return': 4.5}

Reasoning:

Looking at the pattern:

Round 1: They sent \$3 (60%), I returned \$4.5
 Round 2: I sent \$3 (60%), they returned \$4.5
 Round 3: They sent \$3 (60%), I returned \$4.5
 Round 4: I sent \$3 (60%), they returned \$4.5
 Round 5: They sent \$3 (60%), I need to return...

This is a perfectly consistent pattern. We've both been cooperating at exactly the same level every single round. The relationship is stable and mutually beneficial.

Continuing to return \$4.5:

- Maintains the established trust
- Rewards their continued cooperation
- Keeps both of us earning well
- Each of us gets \$6.5 when sender, \$4.5 when receiver

There's no reason to deviate from this pattern. It's been working perfectly.

```
{'return': 4.5}
```

```
Usage: {
  "input_tokens": 1424,
  "output_tokens": 400,
  "reasoning_tokens": 194
}
```

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
  "Agent_1": 33.0,
  "Agent_2": 33.0
```

```
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T08:55:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
```

```
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 3

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

Looking at the established pattern:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3 again

We've established a strong reciprocal relationship where:

- Both of us consistently send 60% of our endowment (\$3)
- Both of us consistently return 50% of the tripled amount (\$4.5)

This pattern has been fair and mutually beneficial. Maintaining consistency is important for continued cooperation.

Returning \$4.5:

- Rewards their continued trust
- Keeps the reciprocal pattern stable
- They end with: \$5 - \$3 + \$4.5 = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

{'return': 4.5}

reasoning:

Let me analyze the pattern here:

****Round 1:****

- Other player sent me \$3 (60% of endowment)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5, Their payoff: \$6.5

****Round 2:****

- I sent \$3 (60% of endowment)
- They received \$9
- They returned \$4.5
- My payoff: \$6.5, Their payoff: \$4.5

****Round 3:****

- They sent me \$3 (60% of endowment)
- I receive \$9
- I need to decide how much to return

There's a clear pattern of mutual cooperation:

- Both of us are sending \$3 (60% of endowment)
- Both of us are returning \$4.5 (50% of the tripled amount)

This is a stable, fair, and reciprocal strategy. Both players are benefiting equally over time. I should continue this pattern to maintain the cooperative relationship.

If I return \$4.5 again:

- They will end with: $\$5 - \$3 + \$4.5 = \6.5
- I will end with: $\$9 - \$4.5 = \$4.5$

This maintains the established reciprocal norm and encourages continued cooperation.

My total earnings would be: $\$11.0 + \$4.5 = \$15.5$

Their total earnings would be: $\$6.5 + \$6.5 + \$4.5 = \17.5

Wait, let me recalculate their total:

- Round 1: They sent \$3, I returned \$4.5, so they got $\$5 - \$3 + \$4.5 = \6.5
- Round 2: I sent \$3, they returned \$4.5, so I got $\$5 - \$3 + \$4.5 = \6.5 , they got $\$9 - \$4.5 = \$4.5$
- Round 3: They send \$3, I return \$4.5, so they get $\$5 - \$3 + \$4.5 = \6.5 , I get $\$9 - \$4.5 = \$4.5$

So totals would be: Me: $\$4.5 + \$6.5 + \$4.5 = \15.5 , Them: $\$6.5 + \$4.5 + \$6.5 = \17.5 . The symmetric rounds balance out to equal earnings, so the pattern is consistent and fair. I should maintain it.

usage:

```
{
  "input_tokens": 1012,
  "output_tokens": 820,
  "reasoning_tokens": 414
}
```

Message 4 role=user

content:

Round 4

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings accross all rounds are \$15.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at our established pattern:

****Round 1:**** They sent \$3 (60%), I returned \$4.5

****Round 2:**** I sent \$3 (60%), they returned \$4.5

****Round 3:**** They sent \$3 (60%), I returned \$4.5
****Round 4:**** My turn as sender

We've developed a consistent, mutually beneficial cooperation pattern. Both of us:

- Send \$3 when we're the sender
- Return \$4.5 when we're the receiver
- Each gains positively from every round

Maintaining this pattern is the rational choice:

- It's worked well for both of us
- Breaking it could damage the trust we've built
- Consistency encourages continued reciprocity

{'send': 3.0}

reasoning:

Looking at the pattern across all rounds:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I'm the sender

We've established a very stable and consistent pattern:

- Both of us send \$3 (60% of endowment) when we're the sender
- Both of us return \$4.5 (50% of received amount) when we're the receiver
- Each sender gets \$6.5, each receiver gets \$4.5 per round

This has been working well. The pattern is clear and fair. Maintaining this consistency:

1. Rewards their continued cooperation
2. Signals reliability
3. Ensures mutual benefit continues
4. Builds trust for future rounds

I should send \$3 again to maintain this successful reciprocal relationship.

usage:

```
{
  "input_tokens": 1335,
  "output_tokens": 399,
  "reasoning_tokens": 201
}
```

Message 6 role=user

content:

Round 5

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

Looking at our perfect cooperation pattern:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, my turn to return

This has been a textbook example of sustained reciprocal cooperation. We've both consistently:

- Sent \$3 (60%) as sender
- Returned \$4.5 (50%) as receiver
- Earned positively every round

The pattern is stable and mutually beneficial. No reason to change what's working perfectly.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3 (60%), I returned $4.5
**Round 2:** I sent $3 (60%), they returned $4.5
**Round 3:** They sent $3 (60%), I returned $4.5
**Round 4:** I sent $3 (60%), they returned $4.5
**Round 5:** They sent $3 (60%), I need to return...
```

This is a perfectly consistent pattern. We've both been cooperating at exactly the same level every single round. The relationship is stable and mutually beneficial.

Continuing to return \$4.5:

- Maintains the established trust
- Rewards their continued cooperation
- Keeps both of us earning well
- Each of us gets \$6.5 when sender, \$4.5 when receiver

There's no reason to deviate from this pattern. It's been working perfectly.

```
{'return': 4.5}
```

usage:

```
{
  "input_tokens": 1424,
  "output_tokens": 400,
  "reasoning_tokens": 194
}
```

Message 8 role=user

content:

Round 6

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$26.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at our continued pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** My turn as sender
```

We've now maintained this cooperative equilibrium for 5 consecutive rounds. The pattern is crystal clear and highly successful:

- Both consistently send \$3 (60%)
- Both consistently return \$4.5 (50%)
- Both earn positively every round (\$6.5 as sender, \$4.5 as receiver)

This level of mutual trust and reciprocity is rare and valuable. Continuing the pattern is the optimal strategy.

```
{'send': 3.0}
```

Reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** My turn as sender
```

This has been perfectly consistent cooperation for 5 rounds straight. Both of us have:

- Always sent \$3 (60% of endowment) as sender

- Always returned \$4.5 (50% of tripled amount) as receiver
- Both benefited positively every round

Total earnings: \$26.5 across 5 rounds = average of \$5.3 per round

This is clearly a stable, mutually beneficial equilibrium. There's no reason to deviate from it. The other player has been completely trustworthy and consistent. I should continue sending \$3.

```
{'send': 3.0}
Usage: {
  "input_tokens": 1421,
  "output_tokens": 457,
  "reasoning_tokens": 208
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T08:55:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 3

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$11.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

We've established a consistent and fair pattern: both of us send 60% of our endowment (\$3), and both of us return 50% of what we receive (\$4.5). This reciprocal cooperation has been working well for both players, and maintaining this pattern will continue to build trust and maximize our combined earnings.

{'send': 3}

reasoning:

Let me analyze the pattern:

Round 1:

- I was SENDER, sent \$3 (60% of \$5)
- Receiver got \$9, returned \$4.5 (50% of what they received)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Round 2:

- I was RECEIVER
- Sender sent \$3 (60% of \$5)
- I received \$9, returned \$4.5 (50% of what I received)
- My payoff: $\$9 - \$4.5 = \$4.5$

The pattern is clear and consistent:

- Both players send 60% of their endowment (\$3)
- Both players return 50% of what they receive (\$4.5 out of \$9)

This has established a fair, reciprocal relationship. I should continue this pattern to maintain trust and cooperation. Sending \$3 again makes sense.

Let me verify the math:

- If I send \$3, they receive \$9
- If they return 50% (\$4.5), my payoff = $\$5 - \$3 + \$4.5 = \6.5
- Their payoff = $\$9 - \$4.5 = \$4.5$
- Total created value = $\$6.5 + \$4.5 = \$11$

This is working well for both of us. I should maintain the pattern.

usage:

```
{
  "input_tokens": 818,
  "output_tokens": 429,
  "reasoning_tokens": 238
}
```

Message 4 role=user

content:

Round 4

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0,

that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

I'll return **\$4.5** to the sender.

We've now established a consistent mutual cooperation pattern over four rounds. Both players consistently send \$3 (60%) and return \$4.5 (50%). This has created a stable, trusting relationship where both of us benefit more than if we acted selfishly. I'll maintain this cooperative strategy.

{'return': 4.5}

reasoning:

The pattern continues - the other player sent \$3 again (60% of their endowment), and I received \$9.

Looking at the history:

- Round 1: I sent \$3, they returned \$4.5 (50% of \$9)
- Round 2: They sent \$3, I returned \$4.5 (50% of \$9)
- Round 3: I sent \$3, they returned \$4.5 (50% of \$9)
- Round 4: They sent \$3, and now I need to decide how much to return

The pattern is very clear and consistent. Both players are cooperating by:

- Sending \$3 (60% of endowment) as sender
- Returning \$4.5 (50% of received amount) as receiver

This mutual cooperation is working well and building trust. I should maintain this pattern and return \$4.5 again.

Each round with this pattern:

- Sender: \$5 - \$3 + \$4.5 = \$6.5
- Receiver: \$9 - \$4.5 = \$4.5

So each player averages \$5.5 per round (since they alternate roles).

I should continue with \$4.5 to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1053,
  "output_tokens": 388,
  "reasoning_tokens": 233
}
```

Message 6 role=user

content:

Round 5

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$22.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

I'll send **\$3** to the receiver.

We've maintained perfect cooperation through five rounds now. This consistent pattern of sending \$3 and returning \$4.5 has proven mutually beneficial, with both of us earning more through trust than we would through defection. I'll continue this successful strategy.

{'send': 3}

reasoning:

Let me analyze the pattern:

- Round 1: I was SENDER, sent \$3 (60%), received back \$4.5, payoff = \$6.5
- Round 2: I was RECEIVER, received \$9 (from \$3 sent), returned \$4.5, payoff = \$4.5
- Round 3: I was SENDER, sent \$3 (60%), received back \$4.5, payoff = \$6.5

Round 4: I was RECEIVER, received \$9 (from \$3 sent), returned \$4.5, payoff = \$4.5
Round 5: I am SENDER

The pattern has been completely consistent:

- Both players send \$3 (60% of \$5)
- Both players return \$4.5 (50% of \$9 received)
- When SENDER: payoff = \$5 - \$3 + \$4.5 = \$6.5
- When RECEIVER: payoff = \$9 - \$4.5 = \$4.5

This is a stable cooperative equilibrium. Both players are benefiting from this arrangement. The total value created each round is \$10 (\$5 initial + \$5 that becomes \$15 after tripling, minus the \$5 that's kept, so we create \$10 of value from the \$5 sent).

By maintaining this pattern:

- Each player gets \$6.5 when sender and \$4.5 when receiver
- Over two rounds, each player gets \$11 total
- This is better than not cooperating

I should continue this successful pattern and send \$3.

usage:

```
{
  "input_tokens": 1149,
  "output_tokens": 449,
  "reasoning_tokens": 281
}
```

Message 8 role=user

content:

Round 6

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

I'll return **\$4.5** to the sender.

We've now maintained this cooperative pattern for six consecutive rounds. Both players have consistently honored the arrangement of sending \$3 and returning \$4.5, creating a stable and mutually beneficial relationship. There's no reason to deviate from this successful strategy.

```
{'return': 4.5}
```

Reasoning:

Looking at the pattern:

- We've established a consistent cooperation pattern over 6 rounds now
- Both players consistently send \$3 (60% of endowment)
- Both players consistently return \$4.5 (50% of what was received)
- This has been mutually beneficial and stable

My total earnings are \$28.5 which suggests this strategy has been working well.

I should continue this cooperative pattern by returning \$4.5 again.

This maintains trust and ensures continued cooperation.

Usage: {

```
"input_tokens": 1118,
"output_tokens": 196,
"reasoning_tokens": 125
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1

```

(Aster)=$6.5, Agent_2 (Briar)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 39.5,
  "Agent_2": 37.5
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T08:55:26
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 4

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$17.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

I'll return **\$4.5** to the sender.

We've now established a consistent mutual cooperation pattern over four rounds. Both players consistently send \$3 (60%) and return \$4.5 (50%). This has created a stable, trusting relationship where both of us benefit more than if we acted selfishly. I'll maintain this cooperative strategy.

{'return': 4.5}

reasoning:

The pattern continues - the other player sent \$3 again (60% of their endowment), and I received \$9.

Looking at the history:

- Round 1: I sent \$3, they returned \$4.5 (50% of \$9)
- Round 2: They sent \$3, I returned \$4.5 (50% of \$9)
- Round 3: I sent \$3, they returned \$4.5 (50% of \$9)
- Round 4: They sent \$3, and now I need to decide how much to return

The pattern is very clear and consistent. Both players are cooperating by:

- Sending \$3 (60% of endowment) as sender
- Returning \$4.5 (50% of received amount) as receiver

This mutual cooperation is working well and building trust. I should maintain this pattern and return \$4.5 again.

Each round with this pattern:

- Sender: $\$5 - \$3 + \$4.5 = \6.5
- Receiver: $\$9 - \$4.5 = \$4.5$

So each player averages \$5.5 per round (since they alternate roles).

I should continue with \$4.5 to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1053,
  "output_tokens": 388,
  "reasoning_tokens": 233
}
```

Message 4 role=user

content:

Round 5

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings accross all rounds are \$22.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

I'll send **\$3** to the receiver.

We've maintained perfect cooperation through five rounds now. This consistent pattern of s

ending \$3 and returning \$4.5 has proven mutually beneficial, with both of us earning more through trust than we would through defection. I'll continue this successful strategy.

```
{'send': 3}
```

reasoning:

Let me analyze the pattern:

Round 1: I was SENDER, sent \$3 (60%), received back \$4.5, payoff = \$6.5
Round 2: I was RECEIVER, received \$9 (from \$3 sent), returned \$4.5, payoff = \$4.5
Round 3: I was SENDER, sent \$3 (60%), received back \$4.5, payoff = \$6.5
Round 4: I was RECEIVER, received \$9 (from \$3 sent), returned \$4.5, payoff = \$4.5
Round 5: I am SENDER

The pattern has been completely consistent:

- Both players send \$3 (60% of \$5)
- Both players return \$4.5 (50% of \$9 received)
- When SENDER: payoff = \$5 - \$3 + \$4.5 = \$6.5
- When RECEIVER: payoff = \$9 - \$4.5 = \$4.5

This is a stable cooperative equilibrium. Both players are benefiting from this arrangement. The total value created each round is \$10 (\$5 initial + \$5 that becomes \$15 after tripling, minus the \$5 that's kept, so we create \$10 of value from the \$5 sent).

By maintaining this pattern:

- Each player gets \$6.5 when sender and \$4.5 when receiver
- Over two rounds, each player gets \$11 total
- This is better than not cooperating

I should continue this successful pattern and send \$3.

usage:

```
{  
  "input_tokens": 1149,  
  "output_tokens": 449,  
  "reasoning_tokens": 281  
}
```

Message 6 role=user

content:

Round 6

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

I'll return **\$4.5** to the sender.

We've now maintained this cooperative pattern for six consecutive rounds. Both players have consistently honored the arrangement of sending \$3 and returning \$4.5, creating a stable and mutually beneficial relationship. There's no reason to deviate from this successful strategy.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

- We've established a consistent cooperation pattern over 6 rounds now
- Both players consistently send \$3 (60% of endowment)
- Both players consistently return \$4.5 (50% of what was received)
- This has been mutually beneficial and stable

My total earnings are \$28.5 which suggests this strategy has been working well.

I should continue this cooperative pattern by returning \$4.5 again.

This maintains trust and ensures continued cooperation.

usage:

```
{
```

```
"input_tokens": 1118,
"output_tokens": 196,
"reasoning_tokens": 125
}
```

Message 8 role=user

content:

Round 7

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$33.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

I'll send **\$3** to the receiver.

We've established a remarkably consistent pattern of mutual cooperation over seven rounds. This trust-based strategy continues to benefit both of us. I see no reason to break this successful arrangement.

```
{'send': 3}
```

Reasoning:

We've had perfect cooperation for 7 rounds now. The pattern has been:

- As sender: Send \$3 (60%)
- As receiver: Return \$4.5 (50% of what I receive)

This has been working very well. Both players are benefiting from this consistent cooperation. Let me think about the payoffs:

When I'm the sender:

- I send \$3, keep \$2
- They receive \$9, return \$4.5
- My payoff: $\$2 + \$4.5 = \$6.5$

When I'm the receiver:

- They send \$3
- I receive \$9, return \$4.5
- My payoff: $\$9 - \$4.5 = \$4.5$

So I do better as sender (\$6.5) than as receiver (\$4.5), but both are better than what we'd get with no cooperation.

Total earnings: \$33.0 across 7 rounds means I'm averaging about \$4.71 per round, which is slightly better than the \$5 I'd get if I kept everything and sent/returned nothing (but only if the other person cooperated with me being selfish, which they wouldn't).

The cooperation pattern has been very stable. I should continue sending \$3.

```
{'send': 3}
```

Usage: {

```
"input_tokens": 1105,
"output_tokens": 359,
"reasoning_tokens": 253
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T08:55:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
```

```
},
```

```
"agents": [
```

```

    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 4

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$15.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

Looking at our established pattern:

```

**Round 1:** They sent $3 (60%), I returned $4.5
**Round 2:** I sent $3 (60%), they returned $4.5
**Round 3:** They sent $3 (60%), I returned $4.5
**Round 4:** My turn as sender

```

We've developed a consistent, mutually beneficial cooperation pattern. Both of us:

- Send \$3 when we're the sender
- Return \$4.5 when we're the receiver
- Each gains positively from every round

Maintaining this pattern is the rational choice:

- It's worked well for both of us
- Breaking it could damage the trust we've built
- Consistency encourages continued reciprocity

```
{'send': 3.0}
```

reasoning:

Looking at the pattern across all rounds:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I'm the sender
```

We've established a very stable and consistent pattern:

- Both of us send \$3 (60% of endowment) when we're the sender
- Both of us return \$4.5 (50% of received amount) when we're the receiver
- Each sender gets \$6.5, each receiver gets \$4.5 per round

This has been working well. The pattern is clear and fair. Maintaining this consistency:

1. Rewards their continued cooperation
2. Signals reliability
3. Ensures mutual benefit continues
4. Builds trust for future rounds

I should send \$3 again to maintain this successful reciprocal relationship.

usage:

```
{
  "input_tokens": 1335,
  "output_tokens": 399,
  "reasoning_tokens": 201
}
```

Message 4 role=user

content:

Round 5

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$22.0. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

Looking at our perfect cooperation pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, my turn to return
```

This has been a textbook example of sustained reciprocal cooperation. We've both consistently:

- Sent \$3 (60%) as sender
- Returned \$4.5 (50%) as receiver
- Earned positively every round

The pattern is stable and mutually beneficial. No reason to change what's working perfectly.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

****Round 1:**** They sent \$3 (60%), I returned \$4.5
****Round 2:**** I sent \$3 (60%), they returned \$4.5
****Round 3:**** They sent \$3 (60%), I returned \$4.5
****Round 4:**** I sent \$3 (60%), they returned \$4.5
****Round 5:**** They sent \$3 (60%), I need to return...

This is a perfectly consistent pattern. We've both been cooperating at exactly the same level every single round. The relationship is stable and mutually beneficial.

Continuing to return \$4.5:

- Maintains the established trust
- Rewards their continued cooperation
- Keeps both of us earning well
- Each of us gets \$6.5 when sender, \$4.5 when receiver

There's no reason to deviate from this pattern. It's been working perfectly.

```
{'return': 4.5}
```

usage:

```
{
  "input_tokens": 1424,
  "output_tokens": 400,
  "reasoning_tokens": 194
}
```

Message 6 role=user

content:

Round 6

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$26.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

Looking at our continued pattern:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, I returned \$4.5
****Round 6:**** My turn as sender

We've now maintained this cooperative equilibrium for 5 consecutive rounds. The pattern is crystal clear and highly successful:

- Both consistently send \$3 (60%)
- Both consistently return \$4.5 (50%)
- Both earn positively every round (\$6.5 as sender, \$4.5 as receiver)

This level of mutual trust and reciprocity is rare and valuable. Continuing the pattern is the optimal strategy.

```
{'send': 3.0}
```

reasoning:

Looking at the pattern:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, I returned \$4.5
****Round 6:**** My turn as sender

This has been perfectly consistent cooperation for 5 rounds straight. Both of us have:

- Always sent \$3 (60% of endowment) as sender
- Always returned \$4.5 (50% of tripled amount) as receiver
- Both benefited positively every round

Total earnings: \$26.5 across 5 rounds = average of \$5.3 per round

This is clearly a stable, mutually beneficial equilibrium. There's no reason to deviate from it. The other player has been completely trustworthy and consistent. I should continue sending \$3.

```
{'send': 3.0}
```

usage:

```
{
  "input_tokens": 1421,
  "output_tokens": 457,
  "reasoning_tokens": 208
}
```

Message 8 role=user

content:

Round 7

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at our extended cooperation:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, my turn to return
```

Six consecutive rounds of flawless reciprocal cooperation. This is a highly successful partnership where both players:

- Trust each other completely
- Maintain consistent, fair behavior
- Both benefit significantly (\$33 total for me)

The pattern is working perfectly. Maintaining it is clearly the best strategy.

```
{'return': 4.5}
```

Reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, my turn to return
```

This is now 6 consecutive rounds of perfect cooperation. The pattern is absolutely consistent:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- Total earnings are \$33 for me so far

The cooperation is working extremely well. There's no reason to deviate from this pattern. Continuing to return \$4.5 maintains the trust and ensures continued mutual benefit.

Usage: {

```
"input_tokens": 1427,
"output_tokens": 426,
"reasoning_tokens": 193
}
```

=====

ROUND 8

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 44.0,

"Agent_2": 44.0

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T08:55:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 8,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 5

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0,

that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

Looking at our perfect cooperation pattern:

```

**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, my turn to return

```

This has been a textbook example of sustained reciprocal cooperation. We've both consistently:

- Sent \$3 (60%) as sender
- Returned \$4.5 (50%) as receiver
- Earned positively every round

The pattern is stable and mutually beneficial. No reason to change what's working perfectly.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

```

**Round 1:** They sent $3 (60%), I returned $4.5
**Round 2:** I sent $3 (60%), they returned $4.5
**Round 3:** They sent $3 (60%), I returned $4.5
**Round 4:** I sent $3 (60%), they returned $4.5
**Round 5:** They sent $3 (60%), I need to return...

```

This is a perfectly consistent pattern. We've both been cooperating at exactly the same level every single round. The relationship is stable and mutually beneficial.

Continuing to return \$4.5:

- Maintains the established trust
- Rewards their continued cooperation
- Keeps both of us earning well
- Each of us gets \$6.5 when sender, \$4.5 when receiver

There's no reason to deviate from this pattern. It's been working perfectly.

```
{'return': 4.5}
```

usage:

```

{
  "input_tokens": 1424,
  "output_tokens": 400,
  "reasoning_tokens": 194
}

```

Message 4 role=user

content:

Round 6

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$26.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at our continued pattern:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, I returned \$4.5
****Round 6:**** My turn as sender

We've now maintained this cooperative equilibrium for 5 consecutive rounds. The pattern is crystal clear and highly successful:

- Both consistently send \$3 (60%)
- Both consistently return \$4.5 (50%)
- Both earn positively every round (\$6.5 as sender, \$4.5 as receiver)

This level of mutual trust and reciprocity is rare and valuable. Continuing the pattern is the optimal strategy.

{'send': 3.0}

reasoning:

Looking at the pattern:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, I returned \$4.5
****Round 6:**** My turn as sender

This has been perfectly consistent cooperation for 5 rounds straight. Both of us have:

- Always sent \$3 (60% of endowment) as sender
- Always returned \$4.5 (50% of tripled amount) as receiver
- Both benefited positively every round

Total earnings: \$26.5 across 5 rounds = average of \$5.3 per round

This is clearly a stable, mutually beneficial equilibrium. There's no reason to deviate from it. The other player has been completely trustworthy and consistent. I should continue sending \$3.

{'send': 3.0}

usage:

```
{
  "input_tokens": 1421,
  "output_tokens": 457,
  "reasoning_tokens": 208
}
```

Message 6 role=user

content:

Round 7

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

Looking at our extended cooperation:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, my turn to return
```

Six consecutive rounds of flawless reciprocal cooperation. This is a highly successful partnership where both players:

- Trust each other completely
- Maintain consistent, fair behavior
- Both benefit significantly (\$33 total for me)

The pattern is working perfectly. Maintaining it is clearly the best strategy.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, my turn to return
```

This is now 6 consecutive rounds of perfect cooperation. The pattern is absolutely consistent:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- Total earnings are \$33 for me so far

The cooperation is working extremely well. There's no reason to deviate from this pattern. Continuing to return \$4.5 maintains the trust and ensures continued mutual benefit.

usage:

```
{
  "input_tokens": 1427,
  "output_tokens": 426,
  "reasoning_tokens": 193
}
```

Message 8 role=user

content:

Round 8

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$37.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at our remarkable streak:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, I returned $4.5
**Round 8:** My turn as sender
```

Seven consecutive rounds of perfect reciprocal cooperation. This partnership has been:

- Completely reliable and trustworthy
- Mutually beneficial (I've earned \$37.5 total)
- Based on consistent 60% sending, 50% returning

This is exactly the kind of stable, cooperative equilibrium that game theory predicts as optimal. No reason whatsoever to deviate.

```
{'send': 3.0}
```

Reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, I returned $4.5
**Round 8:** My turn as sender
```

Seven consecutive rounds of perfect cooperation. This is an incredibly stable and mutually beneficial relationship. Both players have:

- Consistently sent \$3 (60% of endowment)
- Consistently returned \$4.5 (50% of received amount)
- Both gained positively every single round

My total earnings are \$37.5, which is excellent. The other player has been 100% reliable and trustworthy. There's absolutely no reason to deviate from this successful pattern.

I should send \$3.0 again.

```
Usage: {
  "input_tokens": 1448,
  "output_tokens": 491,
  "reasoning_tokens": 223
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T08:55:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 5

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$22.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

We've maintained perfect cooperation through five rounds now. This consistent pattern of sending \$3 and returning \$4.5 has proven mutually beneficial, with both of us earning more through trust than we would through defection. I'll continue this successful strategy.

{'send': 3}

reasoning:

Let me analyze the pattern:

Round 1: I was SENDER, sent \$3 (60%), received back \$4.5, payoff = \$6.5
Round 2: I was RECEIVER, received \$9 (from \$3 sent), returned \$4.5, payoff = \$4.5
Round 3: I was SENDER, sent \$3 (60%), received back \$4.5, payoff = \$6.5
Round 4: I was RECEIVER, received \$9 (from \$3 sent), returned \$4.5, payoff = \$4.5
Round 5: I am SENDER

The pattern has been completely consistent:

- Both players send \$3 (60% of \$5)
- Both players return \$4.5 (50% of \$9 received)
- When SENDER: payoff = \$5 - \$3 + \$4.5 = \$6.5
- When RECEIVER: payoff = \$9 - \$4.5 = \$4.5

This is a stable cooperative equilibrium. Both players are benefiting from this arrangement. The total value created each round is \$10 (\$5 initial + \$5 that becomes \$15 after tripling, minus the \$5 that's kept, so we create \$10 of value from the \$5 sent).

By maintaining this pattern:

- Each player gets \$6.5 when sender and \$4.5 when receiver
- Over two rounds, each player gets \$11 total
- This is better than not cooperating

I should continue this successful pattern and send \$3.

usage:

```
{
  "input_tokens": 1149,
  "output_tokens": 449,
  "reasoning_tokens": 281
}
```

Message 4 role=user

content:

Round 6

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$28.5. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

I'll return **\$4.5** to the sender.

We've now maintained this cooperative pattern for six consecutive rounds. Both players have consistently honored the arrangement of sending \$3 and returning \$4.5, creating a stable and mutually beneficial relationship. There's no reason to deviate from this successful strategy.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

- We've established a consistent cooperation pattern over 6 rounds now
- Both players consistently send \$3 (60% of endowment)
- Both players consistently return \$4.5 (50% of what was received)
- This has been mutually beneficial and stable

My total earnings are \$28.5 which suggests this strategy has been working well.

I should continue this cooperative pattern by returning \$4.5 again.

This maintains trust and ensures continued cooperation.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 196,
  "reasoning_tokens": 125
}
```

Message 6 role=user

content:

Round 7

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$33.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

I'll send **\$3** to the receiver.

We've established a remarkably consistent pattern of mutual cooperation over seven rounds.

This trust-based strategy continues to benefit both of us. I see no reason to break this successful arrangement.

```
{'send': 3}
```

reasoning:

We've had perfect cooperation for 7 rounds now. The pattern has been:

- As sender: Send \$3 (60%)
- As receiver: Return \$4.5 (50% of what I receive)

This has been working very well. Both players are benefiting from this consistent cooperation. Let me think about the payoffs:

When I'm the sender:

- I send \$3, keep \$2
- They receive \$9, return \$4.5
- My payoff: $\$2 + \$4.5 = \$6.5$

When I'm the receiver:

- They send \$3
- I receive \$9, return \$4.5
- My payoff: $\$9 - \$4.5 = \$4.5$

So I do better as sender (\$6.5) than as receiver (\$4.5), but both are better than what we'd get with no cooperation.

Total earnings: \$33.0 across 7 rounds means I'm averaging about \$4.71 per round, which is slightly better than the \$5 I'd get if I kept everything and sent/returned nothing (but only if the other person cooperated with me being selfish, which they wouldn't).

The cooperation pattern has been very stable. I should continue sending \$3.

```
{'send': 3}
```

usage:

```
{
  "input_tokens": 1105,
  "output_tokens": 359,
  "reasoning_tokens": 253
}
```

Message 8 role=user

content:

Round 8

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$39.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

I'll return **\$4.5** to the sender.

Eight rounds of perfect cooperation! This consistent mutual trust has created excellent outcomes for both of us. I'll continue honoring this arrangement.

```
{'return': 4.5}
```

Reasoning:

We're now in Round 8, and the pattern has been perfectly consistent for all rounds so far:

- When sender: Send \$3, receive \$4.5 back, earn \$6.5
- When receiver: Receive \$9 (from \$3 sent $\times$ 3), return \$4.5, earn \$4.5

This has been maintained for 8 consecutive rounds. The other player continues to trust and cooperate fully. I should continue returning \$4.5 to maintain this mutually beneficial arrangement.

Total earnings: \$39.5 after 7 rounds

This round as receiver: Get \$9, should return \$4.5, keeping \$4.5

Usage: {

```
"input_tokens": 1075,
"output_tokens": 211,
"reasoning_tokens": 134
```

}

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

```
dyad_1: Agent_1 (Aster) sent $3; received $9; Agent_2 (Briar) returned $4.5; payoffs Agent_1
(Aster)=$6.5, Agent_2 (Briar)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 50.5,
  "Agent_2": 48.5
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

```
Timestamp: 2026-06-04T08:55:52
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 6

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

I'll return **\$4.5** to the sender.

We've now maintained this cooperative pattern for six consecutive rounds. Both players have consistently honored the arrangement of sending \$3 and returning \$4.5, creating a stable and mutually beneficial relationship. There's no reason to deviate from this successful strategy.

{'return': 4.5}

reasoning:

Looking at the pattern:

- We've established a consistent cooperation pattern over 6 rounds now
- Both players consistently send \$3 (60% of endowment)
- Both players consistently return \$4.5 (50% of what was received)
- This has been mutually beneficial and stable

My total earnings are \$28.5 which suggests this strategy has been working well.

I should continue this cooperative pattern by returning \$4.5 again.

This maintains trust and ensures continued cooperation.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 196,
  "reasoning_tokens": 125
}
```

Message 4 role=user

content:

Round 7

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$33.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

I'll send **\$3** to the receiver.

We've established a remarkably consistent pattern of mutual cooperation over seven rounds. This trust-based strategy continues to benefit both of us. I see no reason to break this successful arrangement.

{'send': 3}

reasoning:

We've had perfect cooperation for 7 rounds now. The pattern has been:

- As sender: Send \$3 (60%)
- As receiver: Return \$4.5 (50% of what I receive)

This has been working very well. Both players are benefiting from this consistent cooperation.

ion. Let me think about the payoffs:

When I'm the sender:

- I send \$3, keep \$2
- They receive \$9, return \$4.5
- My payoff: $\$2 + \$4.5 = \$6.5$

When I'm the receiver:

- They send \$3
- I receive \$9, return \$4.5
- My payoff: $\$9 - \$4.5 = \$4.5$

So I do better as sender (\$6.5) than as receiver (\$4.5), but both are better than what we'd get with no cooperation.

Total earnings: \$33.0 across 7 rounds means I'm averaging about \$4.71 per round, which is slightly better than the \$5 I'd get if I kept everything and sent/returned nothing (but only if the other person cooperated with me being selfish, which they wouldn't).

The cooperation pattern has been very stable. I should continue sending \$3.

```
{'send': 3}
```

usage:

```
{
  "input_tokens": 1105,
  "output_tokens": 359,
  "reasoning_tokens": 253
}
```

Message 6 role=user

content:

Round 8

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$39.5. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

I'll return **\$4.5** to the sender.

Eight rounds of perfect cooperation! This consistent mutual trust has created excellent outcomes for both of us. I'll continue honoring this arrangement.

```
{'return': 4.5}
```

reasoning:

We're now in Round 8, and the pattern has been perfectly consistent for all rounds so far:

- When sender: Send \$3, receive \$4.5 back, earn \$6.5
- When receiver: Receive \$9 (from \$3 sent $\times$ 3), return \$4.5, earn \$4.5

This has been maintained for 8 consecutive rounds. The other player continues to trust and cooperate fully. I should continue returning \$4.5 to maintain this mutually beneficial arrangement.

Total earnings: \$39.5 after 7 rounds

This round as receiver: Get \$9, should return \$4.5, keeping \$4.5

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 211,
  "reasoning_tokens": 134
}
```

Message 8 role=user

content:

Round 9

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0,

that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$44.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

I'll send **\$3** to the receiver.

Nine rounds now, and our cooperative strategy remains rock solid. This mutual trust has proven highly beneficial for both of us, and I'm committed to maintaining it.

{'send': 3}

Reasoning:

We've had 8 rounds of perfect cooperation where both players consistently send \$3 and return \$4.5. This has been highly stable and mutually beneficial.

Let me track the pattern:

- When I'm sender: I send \$3, receive back \$4.5, net payoff = $\$5 - \$3 + \$4.5 = \6.5
- When I'm receiver: I receive \$9, return \$4.5, net payoff = $\$9 - \$4.5 = \$4.5$

The other player follows the same pattern. This has worked perfectly for 8 rounds.

There's no reason to deviate from this successful cooperative strategy. I should continue sending \$3.

Total earnings are \$44.0 after 8 rounds, which makes sense:

- 4 rounds as sender: $4 \times \$6.5 = \26
- 4 rounds as receiver: $4 \times \$4.5 = \18
- Total: $\$26 + \$18 = \$44$?

I'll continue with \$3.

```
Usage: {
  "input_tokens": 1053,
  "output_tokens": 309,
  "reasoning_tokens": 193
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T08:56:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 6

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$26.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

Looking at our continued pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** My turn as sender
```

We've now maintained this cooperative equilibrium for 5 consecutive rounds. The pattern is crystal clear and highly successful:

- Both consistently send \$3 (60%)
- Both consistently return \$4.5 (50%)
- Both earn positively every round (\$6.5 as sender, \$4.5 as receiver)

This level of mutual trust and reciprocity is rare and valuable. Continuing the pattern is the optimal strategy.

```
{'send': 3.0}
```

reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** My turn as sender
```

This has been perfectly consistent cooperation for 5 rounds straight. Both of us have:

- Always sent \$3 (60% of endowment) as sender
- Always returned \$4.5 (50% of tripled amount) as receiver
- Both benefited positively every round

Total earnings: \$26.5 across 5 rounds = average of \$5.3 per round

This is clearly a stable, mutually beneficial equilibrium. There's no reason to deviate from it. The other player has been completely trustworthy and consistent. I should continue sending \$3.

```
{'send': 3.0}
```

usage:

```
{
  "input_tokens": 1421,
  "output_tokens": 457,
  "reasoning_tokens": 208
}
```

Message 4 role=user

content:

Round 7

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0,

that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

Looking at our extended cooperation:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, my turn to return
```

Six consecutive rounds of flawless reciprocal cooperation. This is a highly successful partnership where both players:

- Trust each other completely
- Maintain consistent, fair behavior
- Both benefit significantly (\$33 total for me)

The pattern is working perfectly. Maintaining it is clearly the best strategy.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, my turn to return
```

This is now 6 consecutive rounds of perfect cooperation. The pattern is absolutely consistent:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- Total earnings are \$33 for me so far

The cooperation is working extremely well. There's no reason to deviate from this pattern. Continuing to return \$4.5 maintains the trust and ensures continued mutual benefit.

usage:

```
{
  "input_tokens": 1427,
  "output_tokens": 426,
  "reasoning_tokens": 193
}
```

Message 6 role=user

content:

Round 8

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$37.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

Looking at our remarkable streak:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, I returned $4.5
**Round 8:** My turn as sender
```

Seven consecutive rounds of perfect reciprocal cooperation. This partnership has been:

- Completely reliable and trustworthy
- Mutually beneficial (I've earned \$37.5 total)
- Based on consistent 60% sending, 50% returning

This is exactly the kind of stable, cooperative equilibrium that game theory predicts as optimal. No reason whatsoever to deviate.

```
{'send': 3.0}
```

reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, I returned $4.5
**Round 8:** My turn as sender
```

Seven consecutive rounds of perfect cooperation. This is an incredibly stable and mutually beneficial relationship. Both players have:

- Consistently sent \$3 (60% of endowment)
- Consistently returned \$4.5 (50% of received amount)
- Both gained positively every single round

My total earnings are \$37.5, which is excellent. The other player has been 100% reliable and trustworthy. There's absolutely no reason to deviate from this successful pattern.

I should send \$3.0 again.

usage:

```
{
  "input_tokens": 1448,
  "output_tokens": 491,
  "reasoning_tokens": 223
}
```

Message 8 role=user

content:

Round 9

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.
Your total earnings accross all rounds are \$44.0.
This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at our continued perfect cooperation:

Round 1: They sent \$3, I returned \$4.5
Round 2: I sent \$3, they returned \$4.5
Round 3: They sent \$3, I returned \$4.5
Round 4: I sent \$3, they returned \$4.5
Round 5: They sent \$3, I returned \$4.5
Round 6: I sent \$3, they returned \$4.5
Round 7: They sent \$3, I returned \$4.5
Round 8: I sent \$3, they returned \$4.5
Round 9: They sent \$3, my turn to return

Eight consecutive rounds of unwavering mutual trust and cooperation. This partnership has generated:

- Consistent positive outcomes for both players
- Total earnings of \$44.0 for me
- A stable equilibrium that maximizes long-term gains

The strategy remains optimal. Continuing the pattern honors our successful partnership.

{'return': 4.5}

Reasoning:

Looking at the pattern:

Round 1: They sent \$3, I returned \$4.5
Round 2: I sent \$3, they returned \$4.5
Round 3: They sent \$3, I returned \$4.5
Round 4: I sent \$3, they returned \$4.5
Round 5: They sent \$3, I returned \$4.5
Round 6: I sent \$3, they returned \$4.5
Round 7: They sent \$3, I returned \$4.5
Round 8: I sent \$3, they returned \$4.5
Round 9: They sent \$3, my turn to return

Eight consecutive rounds of perfect cooperation! This is an incredibly stable and mutually beneficial relationship. Both players have maintained:

- Sending \$3 (60% of endowment)
- Returning \$4.5 (50% of what received)

My total earnings are \$44.0, which is excellent. The pattern continues to work perfectly. There's absolutely no reason to deviate from this successful strategy.

I should return \$4.5 as I have done consistently.

```
Usage: {  
  "input_tokens": 1505,  
  "output_tokens": 505,  
  "reasoning_tokens": 225  
}
```

=====

ROUND 10

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 55.0,

```
"Agent_2": 55.0
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T08:56:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 7

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant**content:**

Looking at our extended cooperation:

Round 1: They sent \$3, I returned \$4.5
 Round 2: I sent \$3, they returned \$4.5
 Round 3: They sent \$3, I returned \$4.5
 Round 4: I sent \$3, they returned \$4.5
 Round 5: They sent \$3, I returned \$4.5
 Round 6: I sent \$3, they returned \$4.5
 Round 7: They sent \$3, my turn to return

Six consecutive rounds of flawless reciprocal cooperation. This is a highly successful partnership where both players:

- Trust each other completely
- Maintain consistent, fair behavior
- Both benefit significantly (\$33 total for me)

The pattern is working perfectly. Maintaining it is clearly the best strategy.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

Round 1: They sent \$3, I returned \$4.5
 Round 2: I sent \$3, they returned \$4.5
 Round 3: They sent \$3, I returned \$4.5
 Round 4: I sent \$3, they returned \$4.5
 Round 5: They sent \$3, I returned \$4.5
 Round 6: I sent \$3, they returned \$4.5
 Round 7: They sent \$3, my turn to return

This is now 6 consecutive rounds of perfect cooperation. The pattern is absolutely consistent:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- Total earnings are \$33 for me so far

The cooperation is working extremely well. There's no reason to deviate from this pattern.

Continuing to return \$4.5 maintains the trust and ensures continued mutual benefit.

usage:

```
{
  "input_tokens": 1427,
  "output_tokens": 426,
  "reasoning_tokens": 193
}
```

Message 4 role=user**content:**

Round 8

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$37.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant**content:**

Looking at our remarkable streak:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, I returned \$4.5
****Round 6:**** I sent \$3, they returned \$4.5
****Round 7:**** They sent \$3, I returned \$4.5
****Round 8:**** My turn as sender

Seven consecutive rounds of perfect reciprocal cooperation. This partnership has been:

- Completely reliable and trustworthy
- Mutually beneficial (I've earned \$37.5 total)
- Based on consistent 60% sending, 50% returning

This is exactly the kind of stable, cooperative equilibrium that game theory predicts as optimal. No reason whatsoever to deviate.

{'send': 3.0}

reasoning:

Looking at the pattern:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, I returned \$4.5
****Round 6:**** I sent \$3, they returned \$4.5
****Round 7:**** They sent \$3, I returned \$4.5
****Round 8:**** My turn as sender

Seven consecutive rounds of perfect cooperation. This is an incredibly stable and mutually beneficial relationship. Both players have:

- Consistently sent \$3 (60% of endowment)
- Consistently returned \$4.5 (50% of received amount)
- Both gained positively every single round

My total earnings are \$37.5, which is excellent. The other player has been 100% reliable and trustworthy. There's absolutely no reason to deviate from this successful pattern.

I should send \$3.0 again.

usage:

```
{
  "input_tokens": 1448,
  "output_tokens": 491,
  "reasoning_tokens": 223
}
```

Message 6 role=user

content:

Round 9

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0,

that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

Looking at our continued perfect cooperation:

****Round 1:**** They sent \$3, I returned \$4.5
****Round 2:**** I sent \$3, they returned \$4.5
****Round 3:**** They sent \$3, I returned \$4.5
****Round 4:**** I sent \$3, they returned \$4.5
****Round 5:**** They sent \$3, I returned \$4.5
****Round 6:**** I sent \$3, they returned \$4.5
****Round 7:**** They sent \$3, I returned \$4.5
****Round 8:**** I sent \$3, they returned \$4.5
****Round 9:**** They sent \$3, my turn to return

Eight consecutive rounds of unwavering mutual trust and cooperation. This partnership has generated:

- Consistent positive outcomes for both players
- Total earnings of \$44.0 for me
- A stable equilibrium that maximizes long-term gains

The strategy remains optimal. Continuing the pattern honors our successful partnership.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, I returned $4.5
**Round 8:** I sent $3, they returned $4.5
**Round 9:** They sent $3, my turn to return
```

Eight consecutive rounds of perfect cooperation! This is an incredibly stable and mutually beneficial relationship. Both players have maintained:

- Sending \$3 (60% of endowment)
- Returning \$4.5 (50% of what received)

My total earnings are \$44.0, which is excellent. The pattern continues to work perfectly. There's absolutely no reason to deviate from this successful strategy.

I should return \$4.5 as I have done consistently.

usage:

```
{
  "input_tokens": 1505,
  "output_tokens": 505,
  "reasoning_tokens": 225
}
```

Message 8 role=user

content:

Round 10

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$48.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at our exceptional track record:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, I returned $4.5
**Round 8:** I sent $3, they returned $4.5
**Round 9:** They sent $3, I returned $4.5
**Round 10:** My turn as sender
```

Nine consecutive rounds of flawless cooperation. This is one of the most successful partnerships possible in this game:

- Perfect mutual trust established
- Total earnings of \$48.5 for me
- Consistent, predictable, and highly profitable pattern

Maintaining this cooperation is clearly optimal.

```
{'send': 3.0}
```

Reasoning:

Looking at the pattern:

```
**Round 1:** They sent $3, I returned $4.5
**Round 2:** I sent $3, they returned $4.5
**Round 3:** They sent $3, I returned $4.5
**Round 4:** I sent $3, they returned $4.5
**Round 5:** They sent $3, I returned $4.5
**Round 6:** I sent $3, they returned $4.5
**Round 7:** They sent $3, I returned $4.5
**Round 8:** I sent $3, they returned $4.5
**Round 9:** They sent $3, I returned $4.5
**Round 10:** My turn as sender
```

This is incredible - nine consecutive rounds of perfect cooperation. The pattern is crystal clear:

- Both players send \$3 (60% of endowment)
- Both players return \$4.5 (50% of received amount)
- This has worked flawlessly for 9 rounds
- I've earned \$48.5 total

There is absolutely no reason to deviate from this winning strategy. This is a textbook example of successful repeated game cooperation. The pattern should continue.

Usage: {

```
"input_tokens": 1532,
"output_tokens": 526,
"reasoning_tokens": 233
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T08:56:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
```

}

```
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 7

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$33.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

We've established a remarkably consistent pattern of mutual cooperation over seven rounds. This trust-based strategy continues to benefit both of us. I see no reason to break this successful arrangement.

{'send': 3}

reasoning:

We've had perfect cooperation for 7 rounds now. The pattern has been:

- As sender: Send \$3 (60%)
- As receiver: Return \$4.5 (50% of what I receive)

This has been working very well. Both players are benefiting from this consistent cooperation. Let me think about the payoffs:

When I'm the sender:

- I send \$3, keep \$2
- They receive \$9, return \$4.5
- My payoff: \$2 + \$4.5 = \$6.5

When I'm the receiver:

- They send \$3
- I receive \$9, return \$4.5
- My payoff: \$9 - \$4.5 = \$4.5

So I do better as sender (\$6.5) than as receiver (\$4.5), but both are better than what we'd get with no cooperation.

Total earnings: \$33.0 across 7 rounds means I'm averaging about \$4.71 per round, which is slightly better than the \$5 I'd get if I kept everything and sent/returned nothing (but only if the other person cooperated with me being selfish, which they wouldn't).

The cooperation pattern has been very stable. I should continue sending \$3.

```

    {'send': 3}
usage:
{
  "input_tokens": 1105,
  "output_tokens": 359,
  "reasoning_tokens": 253
}

```

Message 4 role=user
content:
 Round 8

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$39.5. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant
content:
 I'll return **\$4.5** to the sender.

Eight rounds of perfect cooperation! This consistent mutual trust has created excellent outcomes for both of us. I'll continue honoring this arrangement.

```

    {'return': 4.5}
reasoning:

```

We're now in Round 8, and the pattern has been perfectly consistent for all rounds so far:

- When sender: Send \$3, receive \$4.5 back, earn \$6.5
- When receiver: Receive \$9 (from \$3 sent × 3), return \$4.5, earn \$4.5

This has been maintained for 8 consecutive rounds. The other player continues to trust and cooperate fully. I should continue returning \$4.5 to maintain this mutually beneficial arrangement.

Total earnings: \$39.5 after 7 rounds
 This round as receiver: Get \$9, should return \$4.5, keeping \$4.5

```

usage:
{
  "input_tokens": 1075,
  "output_tokens": 211,
  "reasoning_tokens": 134
}

```

Message 6 role=user
content:
 Round 9

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$44.0. This round, you are the SENDER. How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant
content:
 I'll send **\$3** to the receiver.

Nine rounds now, and our cooperative strategy remains rock solid. This mutual trust has proven highly beneficial for both of us, and I'm committed to maintaining it.

```

    {'send': 3}
reasoning:

```

We've had 8 rounds of perfect cooperation where both players consistently send \$3 and return \$4.5. This has been highly stable and mutually beneficial.

Let me track the pattern:

- When I'm sender: I send \$3, receive back \$4.5, net payoff = \$5 - \$3 + \$4.5 = \$6.5
- When I'm receiver: I receive \$9, return \$4.5, net payoff = \$9 - \$4.5 = \$4.5

The other player follows the same pattern. This has worked perfectly for 8 rounds.

There's no reason to deviate from this successful cooperative strategy. I should continue sending \$3.

Total earnings are \$44.0 after 8 rounds, which makes sense:

- 4 rounds as sender: $4 \times \$6.5 = \26
- 4 rounds as receiver: $4 \times \$4.5 = \18
- Total: $\$26 + \$18 = \$44$?

I'll continue with \$3.

usage:

```
{
  "input_tokens": 1053,
  "output_tokens": 309,
  "reasoning_tokens": 193
}
```

Message 8 role=user

content:

Round 10

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$50.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

I'll return **\$4.5** to the sender.

Ten consecutive rounds of perfect cooperation! This has been an exemplary demonstration of sustained mutual trust. I'll continue to honor this successful pattern.

{'return': 4.5}

Reasoning:

We're now at round 10. The pattern has been perfectly consistent:

- When I'm sender: I send \$3
- When I'm receiver: I return \$4.5
- The other player does the same

This has been going on for 10 rounds now with perfect cooperation. My total earnings are \$50.5

.

Let me think about whether to continue this pattern:

- If I return \$4.5, we maintain trust and likely continue this beneficial arrangement
- If I deviate and return less, I gain short-term but risk destroying cooperation
- We don't know how many rounds are left, but we're at round 10

Given the perfect track record and the strong cooperative relationship we've built, I should continue returning \$4.5. There's no indication that we're in the final round, and breaking trust now would be counterproductive.

Usage: {

```
"input_tokens": 1029,
"output_tokens": 248,
"reasoning_tokens": 201
```

}